

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. EE Sem - II Examination, May 2012  
EE - 711 EHVAC Transmission

Date: 26/05/2012, Saturday

Time: 10.00 a.m. to 1.00 p.m

Maximum Marks: 70

## Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

## SECTION – I

- Q-1 Derive the telegraph equation and its constants for transmission line. 7
- Q-2 A. Draw the lattice diagram and wave diagram when a open ended line is charged with a D.C. source. (take attenuation factor as 50 percent) . 5
- B. Derive the equation of surface voltage gradient for 3, 4 and 6 conductors per bundle. From these equations write down the generalized equation for N conductors per bundle. 6
- C. Explain  $L_{50}$  level of audible noise level for transmission lines. How can you say that  $L_5$  level of AN has higher value than  $L_{50}$  level of AN. 3

OR

- Q-2 A. A 735 KV line has the following details: N=no. of subconductors=4,  $d=3.05$  cm, B=bundle spacing=45.72 cm., height H=20 m, phase separation S=14 m in horizontal configuration. By the mangoldt formula, the maximum conductor surface voltage gradient are 20 KV/cm and 18.4 KV/cm for the centre and outer phases respectively. Calculate the SPL or AN in dB(A) at a distance of 30 m along ground from the centre phase. Assume that the microphone is kept at ground level. 5
- B. Prove that for overhead transmission line the velocity of travelling wave is  $3 \times 10^8$  m/s. 4
- C. Find the distance to be travelled by a travelling wave to attenuate to half the original value, if a line conductor having a resistance of  $0.302 \Omega/\text{km}$ , negligible conductance and a surge impedance of  $500 \Omega$ . 5
- Q-3 A. Explain RL circuit closing transient and derive the equation for the doubling effect. Discuss on the result obtained. 7
- B. Explain the current suppression phenomena related to abnormal switching of circuit breaker. 7

OR

- Q-3 A. Explain the capacitive switching transients related to abnormal switching. 7
- B. Explain the successive reflection with lattice diagram, when the travelling wave is reflected and transmitted between three transition point. 7

## **SECTION – II**

- Q -4** Explain commutation failure fault in an Inverter. State the effect of Single & Multi commutation failure in HVDC system. 7
- Q -5 (A)** State the function of smoothening reactor. What are the criteria to be taken into consideration while designing smoothening reactor? 7
- (B)** Explain basic concept of HVDC Circuit Breaker. Giving block diagram, explain its working. 7

**OR**

- Q -5 (A)** Discuss the reactive power requirement of HVDC system in steady state and transient condition. What are the sources of reactive power? 7
- (B)** Give comparison between Series & Parallel Multi-terminal HVDC systems 7
- Q -6 (A)** Describe all types of HVDC Transmission systems. 7
- (B)** Draw the combined converter control characteristics and discuss each part of it in detail. 7

**OR**

- Q -6 (A)** Write a short note on Current Controller and Constant Extinction Angle Controller. 7
- (B)** Under which condition Monopolar Operation of HVDC system is carried out? What are the advantages and disadvantages associated with metallic return Monopolar operation? 7

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